

AI Engineer

Step 6: Deploy AI Models

CareerCompass — Detailed Learning Notes

What this step is about

Separate training from inference and expose a model through a reliable service.

Core Concepts

- A production service should load a versioned model and preprocessing pipeline rather than retraining per request.
- Validate inputs, enforce limits and return predictable errors.
- Measure latency, throughput, memory and model quality.
- Use Flask/FastAPI, containers or managed serving depending on project requirements.
- Monitor drift and performance; retrain based on evidence.

Skills You Should Be Able To Demonstrate

- Export and load a trained model.
- Build an inference API.
- Containerize it.
- Measure latency.
- Log model version safely.
- Add a health endpoint.

Hands-on Project / Practice

- Deploy a small ML model behind a Flask/FastAPI API.
- Add input validation, health checks and a README explaining the model's limitations.

Recommended References

- Machine Learning with Python and Scikit-Learn — <https://www.youtube.com/watch?v=hDKCxebp88A>
- TensorFlow Documentation — <https://www.tensorflow.org/learn>